d How much less likely it is to regain employment with a SUD Int. SUD share of the homeless, Gutwinski et al. (2021) 21.7% 20.2% 3.78 h1 Service flow of homelessness Int. Homelessness band, HUD PiT 2023 and Burt et al. (2001) 0.40–1.00% 0.62% 0.0216 κϕH Mortality when homeless Ext. Meyer et al. (2023) 1.60 n.a. 1.60 κH Employment penalty of homelessness Ext. Desmond and Gershenson (2016) 0.20 n.a. 0.20 c Food stamp maximum allotment Int. SNAP outlays over output 0.42% 0.43% 0.0879 πV Chance of drawing a voucher Int. Recipients over eligible, HUD 2023 26.6% 27.5% 0.0675 T0 Endowment at entry to adulthood (j = 1) Int. Share of homeless aged 25 to 34, HUD PiT 2024 26.0% 26.4% 0.111

Block 3: SMI chain ι SUD multiplier on SMI onset Ext. Danish registers, Nielsen et al. (2017) 2.00 n.a. 2.00 λm Annual chance of recovering from SMI Ext. SMI ten-year recovery rate, ˚ Asbø et al. (2022) 50% n.a. 0.066 ωm Level of the SMI onset profile Int. SMI share of the population, NSDUH 2024 4.82% 4.83% 0.0037 κue

m How much less likely it is to regain employment with SMI Int. SMI share of the homeless, Gutwinski et al. (2021) 25% 24.7% 7.71 μm SUD recovery multiplier for SMI Ext. Dual-diagnosis relapse, Bradizza et al. (2006) 0.82 n.a. 0.82 κϕm Mortality multiplier for SMI Ext. Olfson et al. (2015) 2.50 n.a. 2.50

Block 4: Recovery and treatment ¯πr Baseline SUD recovery rate Ext. SUD ten-year remission rate, Vaillant (1995) 22% n.a. 0.025 μH SUD recovery when homeless Int. Abstinence, housed/homeless, Milby et al. (2005) 1.47 1.45 0.696 g Treatment effect on recovery Ext. Remission on buprenorphine, Paul et al. (2024) 2.00 n.a. 2.00 ¯τ u Utility cost of treatment Int. Treatment take-up, NSDUH 2024 24.5% 24.9% 4.49 ctreat Public cost of a year of treatment Ext. Residential treatment episode, Roebuck et al. (2003) $9,426 n.a. 0.133

Block 5: Externalities ωc Parent's weight on its child Int. Two-adult over lone-parent households' use, less one, NSDUH 2024 17.5% 17.7% 0.085 δd Earnings the child loses because of a parent's SUD Ext. Earnings lost in utero, Nilsson (2017) 0.24 n.a. 0.24

Block 6: Drug market εs Price elasticity of drug supply Ext. Mace et al. (2020) 1.46 n.a. 1.46

Notes: the Model column is the Housing First benchmark. n.a. marks a parameter set from outside the model, which has no moment of its own to reproduce.

Block 1: Who uses drugs. The five parameters in this block set participation and, more importantly, who participates. Neither policy acts on them directly; they decide how much there is for a policy to act on.

θ, the weight on drugs in utility. It is the drug's weight in the bundle, so it mostly sets how much a

household that uses spends on drugs, but a higher weight also makes using at all worth more. As it

rises, use therefore rises on both the extensive and the intensive margin, and with it the

externalities and what either policy has to act on. It is calibrated internally, at 0.17, to aggregate drug spending by users over their aggregate income, 16.0 percent, from retail drug spending in Midgette et al. (2019) and, in NSDUH 2024, the incomes of adults aged 26 and over who use drugs. b, the marginal utility from the first unit of drugs. The marginal value of the first unit of drugs goes

as θ/b, so a larger b makes the first units worth less and a household with little to spend buys none. As b falls, the marginal utility from the first unit rises, and so do overall drug use and the externalities that policy can affect. It targets a participation rate of 8.90 percent, the NSDUH 2024 rate of a drug use disorder of any severity among adults aged 26 and over.26

πd, the chance that a year of use leads to a SUD. This is what makes using risky: the step from drug

use to a disorder. As it rises, the externalities rise, and the moral hazard of Housing First turns an extra voucher into more SUD, which Treatment First then aims to prevent. It targets the share of users who have a SUD, 41.3 percent among adults aged 26 and over in NSDUH 2024. κϕd, increased mortality with a SUD. κϕd sets how much a SUD raises the chance of dying, and mortal-

ity truncates a user's value of life. Because this is an uninsurable risk, a rise in κϕd hits the households with the highest values of life hardest, which reverses the income gradient; as the stylized model of Section 3 shows, this makes it a key parameter. It is calibrated internally, at 4.00, to the income of users relative to the average, a value within the range of the mortality rate ratios of a SUD in the liter- ature: 3.91 in the Danish registers (Plana-Ripoll et al., 2019) and 6.6 among Norwegians in specialist care for a SUD (Heiberg et al., 2018). Users earn 62 percent of the average in the model against 74 percent in the data. א, the flow value of being alive. אis the other half of the mortality mechanism: κϕd sets how much a

SUD raises the chance of dying, and אwhat the life lost is worth. As it rises, dying costs more and use falls. It targets a VSL of 80 times average labour income (Viscusi and Aldy, 2003).27

Block 2: Costs of homelessness and vouchers. Both policies work on this block, from opposite ends: each parameter says how welfare reducing a state is, and so what a voucher and its removal are worth in it.

κue

d , how much less likely it is to regain employment with a SUD. An unemployed household with

a SUD finds work at the rate ¯πue divided by κue

d $>$ 1. As κue

d rises, the expected unemployment spells of households with a SUD lengthen and they are likelier to lose their housing: this is the link from the disorder to the street, and it shapes

how much Treatment First deters. It is calibrated internally to the SUD share of the homeless, 21.7 percent in Gutwinski et al. (2021). h1, the service flow of homelessness. This parameter sets how welfare reducing the outside option is,

which affects the value of life: a household with little to lose from dying faces a smaller mortality cost of using. As h1 rises, a shelter is less welfare reducing, a voucher is worth less and Treatment First has less to act on. It targets a homelessness rate within 0.40 to 1.00 percent of households: the floor is HUD's 2023 PiT count, which counts a single night; the ceiling is the share homeless at some point over a year, from Burt et al. (2001). Its value is 53 percent of the smallest rental (Appendix A), and the model delivers an aggregate homelessness rate of 0.62 percent. $\kappa\phi H$, mortality when homeless. This parameter sets how much more likely a homeless household is to

die. As it rises, the cost of homelessness rises, a voucher is worth more and Treatment First's deterrence is stronger. It is set externally to multiply baseline mortality by 1.60 at every age, the mortality ratio for homeless adults against poor housed adults in Meyer et al. (2023). $\kappa H$, the employment penalty of homelessness. This parameter sets how much likelier a household in

a shelter is to lose its job and how much less likely it is to find one: separations are $1 + \kappa H$ times

26A user in the model is therefore a household whose use of any illicit drug, marijuana among them, amounts to a disorder, mild included. A household with a SUD is one whose disorder is moderate or severe (next item). Use that never reaches a disorder is not modelled.

27No household has a negative value of life in equilibrium, so death is never optimal. Because the value of being alive enters utility additively and utility is concave, the VSL rises faster than consumption, roughly with its square at the model's risk aversion, as in Hall and Jones (2007). Estimates of the income elasticity of the VSL are lower across people at a point in time, 0.5 to 0.6 in Viscusi and Aldy (2003), than over time, 1.5 to 1.7 in Costa and Kahn (2004).

as frequent and job finding $1 - \kappa H$ times as fast. Housing First recovers that employment directly, whereas Treatment First does so only as the fall in SUD lowers unemployment and homelessness over time. As $\kappa H$ rises, homelessness costs more, and both policies have more to act on. It is set externally at 0.20, from Desmond and Gershenson (2016). c, the food stamp maximum allotment. This parameter sets how low consumption can go for the poor-

est households, with no income, assets or voucher, and so where utility is bounded from below. As it rises, the value of being alive rises, mortality costs more and Treatment First has more to deter. It is calibrated internally to food stamp spending over output, 0.42 percent: outlays from the United States Department of Agriculture (USDA) over GDP. $\pi V$ , the chance of drawing a voucher. This is the policy dial itself, and it holds the whole trade-off.

Raising it houses more households, which reduces the childhood homelessness externality, and makes life worth more, which makes Treatment First's deterrence more effective. In

the benchmark it is calibrated internally, at 0.068, so that the share of income-eligible households who hold a voucher is aligned with the 26.6 percent in HUD's Picture of Subsidized Households, 2023. T0, the endowment at entry to adulthood (j = 1). A child who becomes an adult starts adult life with

T0 times average earnings, which sets how likely a young adult is to end up in a shelter in its first years. As it rises, fewer young adults become homeless, and Treatment First affects behaviour less in the child-rearing years, when parents' choices affect their children. Its moment is the share of the homeless aged 25 to 34, 26.0 percent in HUD's 2024 PiT count.

Block 3: SMI chain. This block sets how much further Treatment First reaches beyond the disorder it prevents: a SUD raises the chance of SMI, and SMI lowers employment and raises mortality.

ι, the SUD multiplier on SMI onset. The hinge of the block: without it, SMI is exogenous to everything

a policy does. As it rises, a SUD causes more SMI, and Treatment First's return rises, because preventing the disorder now prevents some SMI too. It is set externally at 2.00, between the hazard ratio of 1.34 for any SUD that Hjorthøj et al. (2018) estimate for the conversion from schizotypal disorder to schizophrenia and the 6.04 for any substance abuse that Nielsen et al. (2017) estimate for the onset of schizophrenia, both in the Danish registers. λm, the annual chance of recovering from SMI. This parameter sets how long an episode of SMI is

expected to last. As it rises, SMI is less persistent, and Treatment First's deterrence matters less, since the SMI spells a SUD can set off are shorter on average. It is set externally at 0.066, the annual hazard that matches the ten-year clinical recovery rate of 50 percent for bipolar disorder in the first-episode psychosis cohort of ˚Asbø et al. (2022). ωm, the level of the SMI onset profile. This parameter scales the chance of developing SMI at every

age, and so how many households have it. As it rises, a SUD's effect on SMI is larger, and Treatment First has more to act on. It is calibrated internally to the share of the population with SMI, 4.82 percent, the prevalence in NSDUH 2024. κue

m , how much less likely it is to regain employment with SMI. The same as κue

d for SMI: a house- hold with SMI finds work at the rate ¯πue divided by κue